



Hugo Vidal

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Physics-trained researcher pursuing a dual degree in Computer Science and Physics, with research experience spanning machine learning theory, reinforcement learning, robotics, and scientific computing. Interested in understanding the principles governing learning systems while developing scalable AI methods for real-world applications.

RESEARCH EXPERIENCE

Enchanted Tools

Paris, France

Reinforcement Learning Research Engineer - Internship (Advisor: Firas Abi-Farraj)

Mar. 2026 – Sep. 2026

- Developed reinforcement-learning-based locomotion and navigation algorithms for a robot rolling on a sphere (ballbot) contributing to the future of the navigation stack of Mirokaï robot.
- Designed high-fidelity simulation environments and training pipelines using NVIDIA Isaac Lab.
- Created policy evaluation and sim-to-real experimentation workflows.

EPFL

Lausanne, Switzerland

Machine Learning Theory - Master Thesis (Advisor: Prof. Florent Krzakala)

Apr. 2025 – Jul. 2025

- Demonstrated experimentally that partially trained fully connected neural networks can learn meaningful representations despite extensive parameter freezing.
- Developed a first-order analytical framework describing feature evolution under constrained training dynamics.
- Built custom PyTorch infrastructure for controlled studies of representation learning.

Gulliver Laboratory

Paris, France

Graduate Researcher (Advisor: Prof. Olivier Dauchot)

Jan. 2025 – Apr. 2025

- Investigated emergent collective behaviors in populations of simple social robots controlled by onboard policies.
- Built the experimental setup for swarm robotic experiments to characterize the relationship between individual learning dynamics and group-level phenomena.
- Developed automated computer-vision pipelines with OpenCV for robot tracking, state reconstruction and behavioral analysis from video recordings.

Greenerwave

Paris, France

R&D Engineer Intern

Jul. 2024 – Dec. 2024

- Designed and experimentally validated multi-state RF resonator prototypes contributing to the next generation of Greenerwave's antenna.
- Applied Random Matrix Theory to characterize electromagnetic field fluctuations in complex dissipative cavities.
- Developed analytical and numerical models for non-Hermitian wave systems.

EDUCATION

Mines Paris – PSL

Paris, France

M.Sc. in Computer Science (Double Degree)

Sep. 2025 – Present

- Specialization: Machine Learning for Complex Systems.
- Selected coursework: Mathematics of Deep Learning, Computer Vision, Modern Natural Language Processing, Generative Modeling, Graph Neural Networks, Statistical Learning.

ESPCI Paris – PSL

M.Sc. in Physics

– Focus on statistical and quantum physics, probability, optimization and scientific computing.

Paris, France

Sep. 2022 – Aug. 2025

Lycée Lakanal

Preparatory Classes for Grandes Écoles (BCPST)

– Ranked among the top 3% nationwide in competitive entrance examinations (5000+ candidates).

Sceaux, France

2019 – 2021

SELECTED PROJECTS

Physics-Informed Variational Autoencoder for Hyperspectral Unmixing (2025)

– Designed a physics-constrained VAE to estimate endmember spectra and abundance maps from hyperspectral satellite imagery.

From N-Grams to GPT (2025)

– Implemented and compared language models ranging from statistical n-grams to Transformer architectures.

Water Scarcity Forecasting Challenge (2025)

– Developed ensemble-learning pipelines combining LightGBM, XGBoost and Explainable Boosting Machines for time-series forecasting.

TECHNICAL SKILLS

Programming: Python, PyTorch, NumPy, Scikit-Learn, MATLAB, LaTeX

Machine Learning: Deep Learning, Reinforcement Learning, Representation Learning, Generative Models, Scientific Machine Learning

Mathematics: Optimization, Probability Theory, Statistical Physics, Random Matrix Theory

Tools: Git, NVIDIA Isaac Lab, CST Studio Suite

Languages: French (Native), English (Fluent)

SELECTED WORKSHOPS

Leman Workshop on Statistical Physics and Machine Learning

Organizers: Florent Krzakala, Lenka Zdeborová

EPFL

May 2025

PSL Week: Statistical Physics and Machine Learning

Organizers: Giulio Biroli, Francis Bach, Bruno Loureiro

ENS Paris

Nov. 2024